

A Monoidal Guarded Command Language

Elena Di Lavore

March 5, 2026

This note defines a monoidal version of the guarded command language and gives it semantics in imperative categories.

Definition 0.1. An *imperative category* is a distributive copy-discard category with a uniform coproduct-trace satisfying $\mathbf{tr}_S(f) \otimes \mathbb{1}_C = \mathbf{tr}_{S \otimes C}(\delta_{S,A;C}^{-R} \circ (f \otimes \mathbb{1}_C) \circ \delta_{S,B;C}^R)$.

The monoidal GCL

The usual guarded command language has three kinds of terms: expressions, predicates and commands. In this context, the predicates serve three roles: testing properties of programs, generating values that satisfy the predicate, and branching the control flow of programs. When restricting to relational semantics, conflating these three roles is appropriate because the sets of morphisms $X \rightarrow 1$, of morphisms $1 \rightarrow X$ and of total morphisms $X \rightarrow 1 + 1$ are all isomorphic. However, this is not true in general. The monoidal guarded command language, then, needs three different kinds of terms to distinguish between the three roles of relational predicates, so it needs two additional kinds of terms: states for generating values and guards for branching the control flow.

The focus of this work is on the semantics, so we keep the syntax simple. We assume that programs have access to a fixed number of variables, N , and that these are all of the same type, X . Of course, one could extend the syntax of the monoidal guarded command language with proper types and their derivations.

Terms are interpreted as morphisms of a certain type in an imperative category $\mathbb{C}$, where we fix an object X and a natural number $N \in \mathbb{N}$. The semantics of an expression is a total and deterministic morphism of type $X^N \rightarrow X$, where X^N denotes the N -fold $\otimes$ -product of X with itself; the semantics of a guard is a total morphism of type $X^N \rightarrow 1 + 1$; the semantics of a predicate is a morphism of type $X^N \rightarrow 1$; the semantics of a state is a morphism of type $1 \rightarrow X^N$; the semantics of a command is a morphism of type $X^N \rightarrow X^N$.

The syntax for each kind of terms is defined inductively starting from a signature.

Definition 0.2. A *signature* is a pair $\Sigma = (\mathcal{G}, \mathbf{ar})$ of a set $\mathcal{G}$ and a function $\mathbf{ar}: \mathcal{G} \rightarrow \mathbb{N}$. The set $\mathcal{G}$ contains the *generators* and the function $\mathbf{ar}$ assigns an arity to them.

Note that we will alternatively use the arity function to specify an arity or a coarity.

We now define the syntax for each kind of terms and the corresponding semantics. We fix four signatures, Σ_E , Σ_G , Σ_P and Σ_S , for expressions, guards, predicates and states, respectively. For each $N \in \mathbb{N}$, we define expressions, guards, predicates, states and commands on N variables; we assume that N is large enough to interpret any generator, that is, $N \geq \mathbf{ar}(g)$ for each $g \in \Sigma_E + \Sigma_G + \Sigma_P + \Sigma_S$.

Expressions. For each $e_0 \in \mathcal{G}_E$, choose a deterministic and total morphism $\mathcal{I}_0(e_0): X^{\text{ar}(e_0)} \rightarrow X$ in $\mathbb{C}$.

$$\begin{array}{lll} e ::= x_i & i \leq N & \pi_i \\ | e_0(e_1, \dots, e_{\text{ar}(e_0)}) & e_0 \in \mathcal{G}_E & \nu_{X^N}^{\text{ar}(e_0)} \circ (\mathcal{I}(e_1) \otimes \dots \otimes \mathcal{I}(e_{\text{ar}(e_0)})) \circ \mathcal{I}_0(e_0) \end{array}$$

We indicate with $\pi_i: X^N \rightarrow X$ the projection on the i -th component and with $\nu_Y^k: Y \rightarrow Y^k$ the morphism that copies the input k times.

Guards. For each $b_0 \in \mathcal{G}_G$, choose a total morphism $\mathcal{I}_0(b_0): X^{\text{ar}(b_0)} \rightarrow 1 + 1$ in $\mathbb{C}$.

$$\begin{array}{lll} b ::= L & \varepsilon_{X^N} \circ \iota_1 \\ | R & \varepsilon_{X^N} \circ \iota_2 \\ | b_0(e_1, \dots, e_{\text{ar}(b_0)}) & b_0 \in \mathcal{G}_G & \nu_{X^N}^{\text{ar}(b_0)} \circ (\mathcal{I}(e_1) \otimes \dots \otimes \mathcal{I}(e_{\text{ar}(b_0)})) \circ \mathcal{I}_0(b_0) \\ | b_1 \wedge b_2 & & \nu_{X^N} \circ (\mathcal{I}(b_1) \otimes \mathcal{I}(b_2)) \circ \text{AND} \\ | \neg b & & \mathcal{I}(b) \circ \sigma_{1,1}^+ \end{array}$$

We indicate with $\iota_1, \iota_2: 1 \rightarrow 1 + 1$ the first and second inclusions, with $\sigma_{A,B}^+: A + B \rightarrow B + A$ the symmetry in $(\mathbb{C}, +, 0)$, and with $\text{AND}: (1 + 1) \otimes (1 + 1) \rightarrow 1 + 1$ the boolean *and* gate, $\text{AND} := \delta_{1,1;1+1}^R \circ (\mathbb{1}_1 + [[\mathbb{1}_1, \mathbb{1}_1], \mathbb{1}_1])$.

Predicates. For each $p_0 \in \mathcal{G}_P$, choose a morphism $\mathcal{I}_0(p_0): X^{\text{ar}(p_0)} \rightarrow 1$ in $\mathbb{C}$.

$$\begin{array}{lll} p ::= \top & \varepsilon_{X^N} \\ | \perp & !_{X^N} \circ i_1 \\ | p_0(e_1, \dots, e_{\text{ar}(p_0)}) & p_0 \in \mathcal{G}_P & \nu_{X^N}^{\text{ar}(p_0)} \circ (\mathcal{I}(e_1) \otimes \dots \otimes \mathcal{I}(e_{\text{ar}(p_0)})) \circ \mathcal{I}_0(p_0) \\ | p_1 \wedge p_2 & & \nu_{X^N} \circ (\mathcal{I}(p_1) \otimes \mathcal{I}(p_2)) \\ | p_1 +_b p_2 & & \underline{\mathcal{I}(b)} \circ [\mathcal{I}(p_1), \mathcal{I}(p_2)] \\ | \check{b} & & \mathcal{I}(b) \circ \pi_1^+ \\ | p[x_i \setminus e] & i \leq N & \nu_{X^N} \circ (\mathcal{I}(e) \otimes \pi_{-i}) \circ \sigma_{(i,1,\dots,i-1,i+1,\dots,N)} \circ \mathcal{I}(p) \end{array}$$

We indicate with $\underline{f}: Y \rightarrow Y + Y$ the morphism $\underline{f} := \nu_Y \circ (f \otimes \mathbb{1}_Y) \circ \delta_{1,1;Y}^R$, for a morphism $f: Y \rightarrow 1 + 1$, with $\pi_1^+: 1 + 1 \rightarrow 1$ the projection on the first component of a coproduct, with $\pi_{-i}: X^N \rightarrow X^{N-1}$ the $\otimes$ -projection on all but the i -th component, and with $\sigma_\rho: X^N \rightarrow X^N$ the permutation in $(\mathbb{C}, \otimes, 1)$ specified by the permutation ρ on N .

States. For each $s_0 \in \mathcal{G}_S$, choose a morphism $\mathcal{I}_0(s_0): 1 \rightarrow X^{\text{ar}(s_0)}$ in $\mathbb{C}$.

$$\begin{array}{lll} s ::= \perp & !_1 \circ i_{X^N} \\ | [u_1, \dots, u_N] s_0 & s_0 \in \mathcal{G}_S & \mathcal{I}_0(s_0) \circ \nu_{X^{\text{ar}(s_0)}}^N \circ (\mathcal{I}(u_1) \otimes \dots \otimes \mathcal{I}(u_N)) \\ | \coprod_{x_{i_1}, \dots, x_{i_k}} s \cdot s_0 & s_0 \in \mathcal{G}_S \wedge \text{ar}(s_0) = k & \mathcal{I}(s) \circ (\pi_{-i_1, \dots, -i_k} \otimes \mathcal{I}_0(s_0)) \circ \sigma_{(\dots, i_1, \dots, i_k)} \\ | s_1 +_b s_2 & & \mathcal{I}(b) \circ [\mathcal{I}(s_1), \mathcal{I}(s_2)] \\ | s \downarrow p & & \mathcal{I}(s) \circ \nu_{X^N} \circ (\mathbb{1}_{X^N} \otimes \mathcal{I}(p)) \\ | s(e/x_i) & i \leq N & \mathcal{I}(s) \circ \nu_{X^N} \circ (\mathcal{I}(e) \otimes \pi_{-i}) \circ \sigma_{(i,1,\dots,i-1,i+1,\dots,N)} \end{array}$$

We indicate with $u_j: X^{\text{ar}(s_0)} \rightarrow X$ expressions on $\text{ar}(s_0)$ variables, with $\pi_{-i_1, \dots, -i_k}: X^N \rightarrow X^{N-k}$ the $\otimes$ -projection on all but the components $i_1, \dots, i_k$, and with $\sigma_{(\dots, i_1, \dots, i_k)}: X^N \rightarrow X^N$ the permutation in $(\mathbb{C}, \otimes, 1)$ that inserts the last k variables between the first $N - K$ in positions $i_1, \dots, i_k$.

Commands. Finally, we define the syntax and semantics of commands. Suppose we also have a signature Σ_C for commands, where symbols have a coarity, $\text{car}: \mathcal{G}_C \rightarrow \mathbb{N}$, together with an arity, $\text{ar}: \mathcal{G}_C \rightarrow \mathbb{N}$. For each $c_0 \in \mathcal{G}_C$, choose a morphism $\mathcal{I}_0(c_0): X^{\text{ar}(c_0)} \rightarrow X^{\text{car}(c_0)}$.

$c ::= \text{skip}$		$\mathbb{1}_{X^N}$
$ \text{ abort}$		$!_{X^N} \circ \mathbb{1}_{X^N}$
$ [u_1, \dots, u_N]c_0(e_1, \dots, e_{\text{ar}(c_0)}) \quad c_0 \in \mathcal{G}_C$		$\nu_{X^N} \circ (\mathbb{1}_{X^N} \otimes (\nu_{X^N}^{\text{ar}(c_0)} \circ (\mathcal{I}(e_1) \otimes \dots \otimes \mathcal{I}(e_{\text{ar}(c_0)})) \circ \mathcal{I}_0(c_0)))$ $\circ \nu_{X^{N+\text{car}(c_0)}}^N \circ (\mathcal{I}(u_1) \otimes \dots \otimes \mathcal{I}(u_N))$
$ c_1 c_2$		$\mathcal{I}(c_1) \circ \mathcal{I}(c_2)$
$ \text{ if } b \text{ then } c_1 \text{ else } c_2$		$\mathcal{I}(b) \circ [\mathcal{I}(c_1), \mathcal{I}(c_2)]$
$ \text{ while } b \text{ do } c$		$\mathbf{tr}_{X^N}([\mathbb{1}_{X^N}, \mathbb{1}_{X^N}] \circ \mathcal{I}(b) \circ (\mathcal{I}(c) + \mathbb{1}_{X^N}))$
$ \text{ assert } p$		$\nu_{X^N} \circ (\mathbb{1}_{X^N} \otimes \mathcal{I}(p))$
$ x_i \leftarrow e \quad i \leq N$		$\nu_{X^N} \circ (\mathcal{I}(e) \otimes \pi_{-i}) \circ \sigma_{(i, 1, \dots, i-1, i+1, \dots, N)}$
$ (x_{i_1}, \dots, x_{i_k}) \overset{\$}{\leftarrow} s \quad i_j \leq N$		$(\pi_{-i_1, \dots, -i_k} \otimes \mathcal{I}(s)) \circ \sigma_{(\dots, i_1, \dots, i_k)}$

We indicate with $u_i: X^{N+\text{car}(c_0)} \rightarrow X$ expressions on $N + \text{car}(c_0)$ variables.

Future work. It would be good to add the typed version with its derivation rules, at least in the appendix. I should instantiate the monoidal GCL to recover the usual GCL and its probabilistic version, the pGCL. I should also improve typesetting.

Acknowledgements. This note is part of joint work with Mario, Filippo and Sam.